# Arnis 3.4.0 and Confluence.4

What differs, and what it costs on a large area. Confluence.4 is a test build from the fork, not an official Arnis release.

## 1. Features

| Feature | Arnis 3.4.0 | Confluence.4 |
|---|---|---|
| One World mode (many regions, one world folder) | yes | yes |
| Build in pieces with parallel workers (`--unit-regions`, `--one-world-workers`) | no | yes |
| Auto piece size and worker count from RAM and CPU | no | yes |
| RAM budget and thread limit (`--ram-budget-mb`, `--threads`) | no | yes |
| Resume a job, rebuild chosen pieces (`--skip-built`, `--only-pieces`) | no | yes |
| Piece plan on the map, with time and RAM estimate | no | yes |
| Map legend (? button) | no | yes |
| Projects tab: open a world, expand it | no | yes |
| Download now (fill the caches, build nothing) | no | yes |
| Offline mode, local tile archive, Region Download | no | yes |
| Water detail, river bed, climate mode, seed, height limits | no | yes |
| Rocks, bushes, tree packs, props | no | yes |
| Cave style and cave ores | no | yes |
| Road detail, no-buildings, chest loot table | no | yes |
| B_Linear region format, world border, origin | no | yes |
| Polygon Area shapes for selecting land | no | yes (from upstream Arnis) |
| Multi-area merge | no | yes (from upstream Arnis) |
| Discord status | no | yes (from upstream Arnis) |
| Parallel path for Bedrock output | no | yes (from upstream Arnis) |
| Faster Java NBT writer | no | yes (from upstream Arnis) |

With Advanced features off, the world matches stock Arnis main on the test area (0 of 840 files differ). Thirteen window controls from the 3.4.0 fork are not back; their flags still work on the command line.

## 2. Speed and RAM on three areas

Stock 3.4.0 against the tuned Auto setting (`--one-world --unit-regions auto --one-world-workers auto`). Ratio = tuned Auto against stock; a minus sign means tuned Auto is lower. No run timed out or ran out of memory.

| Area | Wall time, median of 3 | | | Peak RAM | | | World size | |
|---|---|---|---|---|---|---|---|---|
| | Stock | Tuned Auto | Ratio | Stock | Tuned Auto | Ratio | Stock | Tuned Auto |
| S8 (64 km²) | 15.3 s | 16.3 s | +6.5% | 2.5 GB | 2.5 GB | +0.4% | 1.0 GB | 1.0 GB |
| S16 (256 km²) | 57.9 s | 52.8 s | -8.8% | 7.2 GB | 3.4 GB | -53.1% | 3.8 GB | 3.9 GB |
| S24 (576 km²) | 94.4 s | 103.1 s | +9.2% | 7.9 GB | 4.8 GB | -39.0% | 8.6 GB | 8.8 GB |

Pieces trade a little speed for much lower RAM. Stock is faster at 64 km² (15.3 s against 16.3 s; tuned Auto builds this size as one run) and at 576 km² (94.4 s against 103.1 s). At 256 km² tuned Auto took 52.8 s against 57.9 s, with 3.4 GB peak RAM against 7.2 GB (-53.1%); at 576 km² the RAM is 4.8 GB against 7.9 GB (-39.0%). Pieces write a slightly larger world (+2.4% at 256 km²).

## Sweep: piece size and worker count

| Regions per piece side / workers | 2 workers | 4 workers | 6 workers |
|---|---|---|---|
| **2×2** | 140.3 s (137.9–141.1)<br>0.7 GB | 101.9 s (100.8–103.0)<br>1.1 GB | 94.8 s (94.7–146.3)<br>1.4 GB |
| **4×4** | 83.3 s (82.4–85.6)<br>1.3 GB | 68.2 s (66.6–71.3)<br>2.1 GB | 71.0 s (70.1–72.2)<br>2.6 GB |
| **8×8** | 71.1 s (67.8–72.1)<br>2.5 GB | 64.9 s (64.8–67.2)<br>3.2 GB<br>**Recommended** | 72.0 s (69.7–74.0)<br>4.3 GB |

Sweep on 256 km² (S16), median of 3 (min–max) and peak RAM, run before the tune. The best median is 64.9 s (8×8 pieces, 4 workers); the next, 4×4 with 4 workers, is 68.2 s, outside the best cell's range, so the tie rule does not apply and the fastest cell wins. Tuned Auto matches the best sweep result only at 256 km²: 52.8 s against 52.8 s for the fixed cell, both in the second session.

These results come from one dense city on one machine. A rural area, a slower disk or less RAM can move them.

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Machine: 24 threads, 31 GB RAM, Windows 11; one machine, three areas. Runs on 2026-11-10 and 2026-11-11. Stock = Arnis 3.4.0 exe (sha256 aafa66e9…). Confluence.4 = branch tip build a78668b6 (sweep), plus the tuned Auto build (branch c35/c4-tune, commit 1cefd515, sha256 b295e96b…) for the tuned numbers. Caches warm; the two cold-cache runs (one each) are in docs/23-BENCH-PIECES-VS-SINGLE.md. Wall time is the median of 3 runs; the grid also shows min–max. The second session ran about 20% faster than the first (stock S16 72.7 s, then 57.9 s), so the tuned comparison uses stock reruns from the same session. Failed, timed-out or out-of-memory runs: 0. Source: docs/bench/results*.jsonl, built by make_pdf.py.